CSA0704 — Computer Networks

Design and Performance Evaluation of a Resilient Smart Campus Network using Modern Layered Protocols

Technical Report

Computer Networks Team Assignment (Marks: 100)

Team Members

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Course Code&Name: CSA0704 – Computer Networks

Assignment Title:	**Design and Performance Evaluation of a Resilient Smart Campus Network using Modern Layered Protocols**
Course Outcome(s) – CO:	**CO1: Apply the functions of the OSI layer in enabling reliable data transmission across networks** **CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using a simulation tool, for efficient network design and topology** **CO3: Analyze and compare the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions** **CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards** **CO5: Measure network performance using key metrics with simulation tools**
Bloom's Taxonomy Level	**L3 – Apply; L4 – Analyse**
SDG Mapping (SDG 1-17)	**SDG 4 – Quality Education** **SDG 9 – Industry, Innovation and Infrastructure** **SDG 11 – Sustainable Cities and Communities**
Team Size	**4 members (same problem statement for all teams)**
Maximum Marks	**100**

Problem Statement

A large multi-campus university is upgrading its network infrastructure to support high-density Wi-Fi, IoT sensors for energy monitoring, online teaching platforms, research data transfers, and administrative services. The existing network suffers from frequent collisions in wireless zones, inefficient IP address allocation, slow convergence of routing tables during link failures, TCP congestion collapse during peak lecture hours, and unreliable application services (DNS resolution delays, email delivery failures, and unsecured file transfers).

Your team of 4 members is required to design, implement (via simulation/emulation), analyse and evaluate a complete end-to-end network solution that addresses the above challenges while incorporating modern technologies from the Data Link, Network, Transport and Application layers.

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Executive Summary

This report presents the design, simulation and performance evaluation of a resilient network for a large multi-campus university. The existing network suffers from wireless collisions, inefficient IP addressing, slow routing convergence, and unreliable application services. The proposed solution uses a three-tier hierarchical topology (core–distribution–access), a hierarchical VLSM-based IPv4 addressing plan, OSPF interior routing supplemented by an SDN overlay, DiffServ QoS, and a defence-in-depth security model from Layer 2 through TLS 1.3.

The design targets 5,000 concurrent wireless users and 500 IoT endpoints, sub-Gbps backbone throughput with under 1% wireless packet loss, DNS resolution under 50 ms, and routing/spanning-tree re-convergence within 2 RTT of a single link failure. The team implemented and validated the design in Cisco Packet Tracer, analysed protocol behaviour with Wireshark, and version-controlled all artefacts in GitHub.

The remainder of this report is organised to mirror the assessment rubric: Section 1 addresses problem understanding, error control and data-link design (CO1); Section 2 covers IP addressing, subnetting and routing (CO2); Section 3 analyses the transport layer, congestion control and QoS (CO3); Section 4 details application-layer protocols and security (CO4); Section 5 documents the modern tools used and simulation evidence (CO1–CO3); Section 6 presents results, trade-off analysis and engineering justification (CO2/CO3); Section 7 discusses broader considerations; Section 8 covers technical documentation and team contribution; Section 9 is the team reflection; and Section 10 gives references and appendices.

1. Problem Understanding, Error Control&Data-Link Design (CO1)

1.1 Restated Problem Statement

The university operates a multi-campus footprint combining teaching zones, research laboratories, administrative offices, residential halls and an outdoor IoT sensor deployment for energy monitoring. The as-is network exhibits four classes of defect that this design must remove:

- **High-density wireless collisions:** frequent CSMA/CA contention and hidden-node collisions in lecture halls and libraries during peak enrolment hours degrade throughput and increase retransmissions.
- **Inefficient address space use:** flat or poorly planned IPv4 allocation wastes addresses, complicates access-control policy, and prevents clean route summarisation.
- **Slow convergence:** a distance-vector-only routing design and unmanaged Layer 2 loops cause multi-second route recovery and broadcast storms after a single uplink failure.
- **Unreliable application services:** web portals, research file transfer and university e-mail see inconsistent latency and no consistent transport-layer congestion behaviour, and are not adequately secured against interception.

These four defects map directly onto the four functional layers the brief requires the team to address: the data-link layer (error detection/correction and medium access), the network layer (addressing and routing), the transport layer (reliability, congestion control and QoS) and the application layer (protocols and security).

1.2 Requirements Traceability

Table 1 restates the functional and non-functional requirements from the assignment brief and links each to the design element that satisfies it; this matrix is revisited in Section 6 when results are evaluated against target values.

ID	Requirement	Type	Design Element Addressing It
R1	≥5,000 concurrent wireless users	Functional	Wi-Fi 6 (802.11ax), OFDMA, band steering, AP density plan (§1.4)
R2	500 IoT devices	Functional	Dedicated IoT VLAN, Target Wake Time, lightweight security (§3.4, 5.4)
R3	Reliable e-mail, web, data transfer	Functional	HTTPS/TLS 1.3, IMAPS, SFTP application design (§5)
R4	Packet loss<1% in wireless zones	Performance	RF/AP planning, QoS queuing, RTS/CTS tuning (§1.4, 4.4)
R5	DNS resolution < 50 ms (local)	Performance	Local recursive DNS caching resolvers per campus (§ 5.2)
R6	TCP recovery within 2 RTT of link failure	Performance	OSPF fast convergence + SDN backup paths (§4.2, 7.1)
R7	Hierarchical addressing, ≥1 modern WLAN standard	Technical	Three-tier hierarchy, VLSM plan, Wi-Fi 6 (§1.4, 3.1)

ID	Requirement	Type	Design Element Addressing It
R8	Wired/wireless error correction/detection	Technical	CRC-32, FCS + block ACK, Hamming-code discussion (§1.3)
R9	SDN-controlled segment, ≥2 layers	Team	SDN/OpenFlow overlay at distribution layer (§3.3)

Table 1. Requirements traceability matrix.

1.3 Data-Link Layer Design: Error Control and Medium Access

1.3.1 Error detection and correction strategy

Two error-control regimes are used, matched to the medium. **Wired Ethernet segments** carry a 32-bit Frame Check Sequence computed with the CRC-32 polynomial, selected over parity or a simple checksum because it detects all single- and double-bit errors and virtually all burst errors up to 32 bits — a good match for the low but non-zero bit-error rate of copper/fibre campus links; CRC runs in switch/NIC hardware, adding no measurable latency.

Wireless segments (802.11ax) also carry an FCS, but because the channel has a materially higher bit-error rate, 802.11 additionally uses immediate/Block Acknowledgement at the MAC layer, so a receiver failing the FCS check simply withholds the ACK, triggering retransmission by the sender rather than a full upper-layer retransmit (further analysed as the CRC-vs-Hamming-vs-ARQ trade-off in Section 6.3).

Scheme	Detects	Corrects	Overhead	Verdict
Parity bit	Odd-count bit errors only	No	Minimal (1 bit)	Rejected — insufficient burst-error coverage for campus link BER
Internet checksum (16-bit)	Most errors, weak on reordering	No	Low	Used only at IP/TCP/UDP headers per RFC 791/793, not link layer
Hamming(7,4)	2-bit errors	1-bit errors	High (~75%)	Rejected for Ethernet — CRC is cheaper and more capable; kept as teaching contrast
CRC-32 (Ethernet FCS)	Single/double-bit + bursts ≤32 bits	No (discard/retransmit)	Low (4 B/frame)	Selected — all wired frames
FCS + Block ACK (802.11)	As CRC-32, plus loss via missing ACK	No (ARQ retransmit)	Low, amortised over A-MPDU	Selected — all wireless frames

Table 2. Data-link error-control scheme comparison and justification.

1.3.2 Medium access control

Wired access-layer ports operate in full-duplex switched Ethernet, so CSMA/CD collision handling does not apply; each port is an isolated collision domain. In the wireless zones, 802.11ax CSMA/CA with OFDMA is used: OFDMA subdivides a channel into resource units so the AP can schedule multiple stations within one transmission opportunity, directly reducing contention-driven collisions in high-density lecture halls and libraries. RTS/CTS is enabled selectively on APs serving rooms with more than 40 associated stations, since it adds overhead not justified in lightly loaded zones.

1.4 Physical and Data-Link Topology

A three-tier hierarchical topology (core–distribution–access) is used rather than a flat or full-mesh design, because it localises broadcast domains, keeps the spanning-tree diameter shallow, and lets the team apply the SDN overlay at a single well-defined layer (the distribution layer) as required by R9.

- **Core layer:** two Layer-3 switches in an HSRP active/standby pair, connected by 10 Gbps fibre EtherChannel, providing the campus backbone and a single point of route summarisation between buildings.
- **Distribution layer:** one Layer-3 switch per zone (Teaching, Research, Admin, Residential, IoT/Sensor), each dual-homed to both core switches for resilience, terminating inter-VLAN routing and hosting the OpenFlow agent for the SDN-controlled segment.
- **Access layer:** Layer-2 switches with port security and 802.1Q trunking to distribution, plus a Wireless LAN Controller managing lightweight 802.11ax access points per building.

Rapid PVST+ is enabled on all Layer-2 links with load-balanced root placement across the two core switches, and PortFast + BPDU Guard are enabled on all access ports to prevent accidental loop formation from patched-in devices — directly addressing the broadcast-storm defect noted in Section 1.1.

2. IP Addressing, Subnetting&Routing Design (CO2)

2.1 Addressing Philosophy

The campus uses the private address block 10.0.0.0/8 (RFC 1918), giving ample room for hierarchical, summarisable allocation across five functional zones — Teaching, Research, Administration, Residential and outdoor IoT/Sensor — plus a data-centre and out-of-band management block. Addressing is assigned top-down: each zone receives a /19 block at the distribution layer, and within a zone each building or floor receives a /22 block, subnetted with Variable-Length Subnet Masking (VLSM) to fit actual host counts rather than a fixed mask. This avoids the address waste identified in Section 1.1 and keeps each zone summarisable into a single core advertisement, bounding the core routing table regardless of how many access subnets exist underneath.

2.2 Campus-Wide Zone Allocation

Zone / Block	CIDR	Usable Hosts	Purpose
Core backbone P2P links	10.0.0.0/24	up to 64 × /30 links	Core–distribution and core–core routed links
Teaching Zone	10.10.0.0/19	8 × /22 buildings	Lecture halls, libraries, computer labs
Research Zone	10.20.0.0/19	8 × /22 buildings	Research labs, HPC cluster access
Administration Zone	10.30.0.0/19	8 × /22 buildings	Registrar, finance, HR, faculty offices
Residential Zone	10.40.0.0/19	8 × /22 halls	Student residence Wi-Fi and wired ports
IoT / Outdoor Sensor Zone	10.50.0.0/19	8 × /22 sub-zones	Energy meters, environmental sensors, smart lighting
Data Centre / Server Farm	10.60.0.0/22	1,022	Application, DNS, DHCP, mail, file servers
DMZ (public-facing)	10.90.0.0/24	254	Web portal, VPN concentrator, reverse proxy
Out-of-Band Management	10.99.0.0/24	254	Device management, SNMPv3, syslog, NTP

Table 3. Campus-wide zone-level address allocation (summarised at the core).

2.3 Worked VLSM Example — Teaching Building A (10.10.0.0/22)

To demonstrate the subnetting methodology applied uniformly across every building, Table 4 works through VLSM allocation for a representative building using its assigned 10.10.0.0/22 block (1,024 addresses). Subnets are allocated largest-first — the standard VLSM discipline for minimising fragmentation — sized from room capacity and device inventory.

VLAN / Segment	Hosts Needed	Mask	Network	Usable Range
20 – Student Wireless	500	/23 (510)	10.10.0.0	10.10.0.1 – 10.10.1.254
10 – Staff Wired	40	/26 (62)	10.10.2.0	10.10.2.1 – 10.10.2.62
50 – IoT Sensors	60	/26 (62)	10.10.2.64	10.10.2.65 – 10.10.2.126
30 – Voice (VoIP)	20	/27 (30)	10.10.2.128	10.10.2.129 – 10.10.2.158
40 – Printers/Peripherals	10	/28 (14)	10.10.2.160	10.10.2.161 – 10.10.2.174
99 – Local Management	6	/29 (6)	10.10.2.176	10.10.2.177 – 10.10.2.182

VLAN / Segment	Hosts Needed	Mask	Network	Usable Range
P2P – Uplink to Distribution	2	/30 (2)	10.10.2.184	10.10.2.185 – 10.10.2.186

Table 4. VLSM subnetting for Teaching-A (700 of 1,024 addresses used; 324 reserved for growth).

The seven subnets consume 700 of 1,024 addresses, leaving 324 unallocated within the /22 for future VLANs without renumbering any existing subnet. The same seven-VLAN template is replicated across the remaining Teaching and Research buildings; a lighter four-VLAN variant (Wired, Wireless, Management, Uplink) is used in Administration and Residential buildings, worked in full in Appendix B.

2.4 VLAN Design

VLAN ID	Name	Typical Zones	Notes
1	Default (unused)	—	No end devices assigned, per hardening best practice
10	Staff-Wired	All	802.1X authenticated wired ports
20	Student/Guest-Wireless	Teaching, Residential	WPA3-Enterprise SSID, captive portal for guests
30	Voice	All	DSCP EF marking, dedicated queue
40	Peripherals	All	Printers, scanners, digital signage
50	IoT/Sensors	IoT zone + building sub-VLANs	Isolated, rate-limited, default-deny east-west
60	Servers	Data Centre	Application, DNS, DHCP, mail, file servers
90	DMZ	DMZ	Reverse proxy, VPN concentrator
99	Management	All (OOB)	Native on management trunk only

Table 5. Campus VLAN scheme (consistent numbering across all distribution zones).

Inter-VLAN routing is performed at the distribution layer using Switch Virtual Interfaces (SVIs), one per VLAN per building block, each configured as the default gateway for its subnet and, where redundancy is required (Servers, DMZ, core-adjacent Staff-Wired), backed by HSRP with active/standby roles alternated across the two core switches to balance forwarding load.

2.5 Routing Design

2.5.1 Interior gateway protocol selection

OSPFv2 (RFC 2328) is used as the interior gateway protocol, configured as a multi-area design (Area 0 on core links, one non-backbone area per zone) with each zone's distribution router acting as Area Border Router (ABR). OSPF was selected over EIGRP and RIP for three reasons directly relevant to Section 1.1's defect list: **convergence speed** — link-state SPF recomputation plus Bidirectional Forwarding Detection (BFD) drops failure detection to sub-second, targeting R6; **scalability via areas** — an outage in one zone's area does not trigger campus-wide SPF recomputation; and **vendor neutrality** — unlike Cisco-originated EIGRP, OSPF is a fully open IETF standard suited to multi-vendor procurement. EIGRP is retained only on one legacy Residential segment for an empirical DUAL-vs-SPF convergence comparison (Section 6.3), redistributed into OSPF with route tags to prevent loops.

2.5.2 Route summarisation

Each ABR summarises its zone's /22 building subnets into the single /19 zone block (Table 3) before advertising into Area 0, bounding the core routing table to nine entries regardless of building or VLAN count underneath, keeping SPF recomputation fast as the campus grows.

2.5.3 SDN-controlled segment

To satisfy the requirement for at least one SDN-controlled segment with policy-based routing, links between the core pair and the five distribution switches are instrumented as an OpenFlow-managed overlay under a central SDN controller, maintaining a shortest-latency and a highest-bandwidth pre-computed path per distribution switch, steerable via flow rules keyed on DSCP marking (Section 4.4). This layer sits above OSPF (which guarantees reachability) and gives near-zero-delay failover for pre-programmed backup paths, satisfying R9's two-layer division of responsibility.

2.5.4 DHCP and address assignment

End-host addressing is dynamic via DHCP, with one relay agent per SVI (ip helper-address) pointing to the centralised Data Centre DHCP service rather than a per-building server, keeping pool administration consistent with the summarised addressing plan. Each VLAN's scope reserves the first ten addresses of its usable range for static infrastructure assignment.

2.6 IPv6 Note

The design reserves a Unique Local Address prefix, fd00:campus::/48 (one /56 per zone), so dual-stack can be enabled zone-by-zone in a future phase without renumbering the IPv4 plan. IPv6 is not implemented or simulated in the current topology — noted as a scope boundary and limitation in Section 6.4.

3. Transport Layer, Congestion Control&QoS Analysis (CO3)

3.1 Transport Protocol Selection per Application

Each campus service is assigned to TCP or UDP based on whether it needs reliable, ordered delivery or low latency with tolerance for occasional loss, rather than defaulting every service to TCP as the legacy network did.

Application	Transport	Port(s)	Rationale
Web portal / LMS (HTTPS)	TCP	443	Ordered, reliable delivery required for the TLS byte stream
Research file transfer (SFTP)	TCP	22	Large datasets must not be corrupted or reordered
Staff/student e-mail (IMAPS)	TCP	993	Mailbox state and message integrity require reliability
DNS (recursive queries)	UDP (TCP fallback)	53	Single request/response; TCP only for zone transfers
Campus VoIP	UDP (RTP/RTCP)	16384–32767	Timeliness matters more than occasional lost samples
SNMPv3 monitoring	UDP	161/162	Lightweight polling; a lost poll is simply retried
IoT sensor telemetry	UDP (CoAP-style)	5683	High-volume small payloads; TCP overhead disproportionate

Table 6. Transport-layer protocol assignment per application.

3.2 TCP Congestion Control Behaviour

All TCP flows use a Cubic-family congestion control algorithm at the end hosts (the OS default), which the network assumes rather than re-implements. Four phases form the analytical baseline for Section 6.1: **slow start** — cwnd begins at the RFC 6928 initial window and doubles each RTT until ssthresh; **congestion avoidance** — beyond ssthresh, cwnd grows linearly; **fast retransmit** — three duplicate ACKs trigger immediate retransmission, avoiding a costly retransmission-timeout on a well-provisioned LAN; and **fast recovery** — ssthresh is halved and cwnd set near the new ssthresh, which is what makes the R6 target (recovery within 2 RTT of a single loss) realistic. MSS is 1460 bytes from the 1500-byte Ethernet MTU; the WLAN's effective MTU is kept at 1500 bytes at the 802.11 MAC-SDU boundary to avoid fragmentation across the wireless hop.

3.3 UDP and Real-Time Traffic Handling

Because UDP has no congestion control of its own, RTP (VoIP) flows are rate-shaped at the source (20 ms packetisation, G.711/Opus-class bit rates) and never exceed their provisioned share, so the absence of TCP-style backoff does not risk starving other traffic — managed instead by the DiffServ scheduling in Section 3.4. IoT telemetry is similarly rate-limited per device at the access switch so a malfunctioning sensor cannot flood its VLAN.

3.4 Quality of Service (QoS) Design

A DiffServ model is used rather than IntServ, because IntServ's per-flow reservation state does not scale to 5,000+ concurrent wireless clients, whereas DiffServ classifies and marks traffic once at the network edge and needs only aggregate per-class queuing in the core.

Class	Traffic	DSCP	Queuing Treatment	Target
Voice	VoIP RTP	EF (46)	Strict-priority (LLQ)	<30 ms delay, <1% loss, <5 ms jitter
Signalling	SIP/SCCP call control	CS3 (24)	Priority, rate-limited	Low delay, no call-setup loss
Video/interactive	Video-conf, remote labs	AF41 (34)	WFQ, guaranteed min. bandwidth	<150 ms delay, <1% loss
Critical data	SNMP, DNS, DHCP, OSPF/BFD	CS6 (48)	Priority (control-plane protection)	Never starved by bulk data
Transactional	Web/HTTPS, IMAPS	AF21 (18)	WFQ, moderate guaranteed share	Responsive interactive experience
Bulk data	SFTP transfers, backups	AF11/Default	WFQ, remainder of bandwidth, WRED	High goodput, tolerant of delay
IoT telemetry	Sensor uploads	CS1 (8)	Rate-limited, best-effort	Low priority, bounded share

Table 7. DiffServ traffic classes, DSCP marking and queuing treatment.

Marking is trusted only from the WLC, IP phones and the SDN controller's own control channel; all other ingress marking is re-classified at the access switch (Section 4.4) so end-user devices cannot self-mark into the priority queue. Weighted Random Early Detection (WRED) on the bulk-data queue gives early, distributed drop signals as it fills, which — combined with fast retransmit/fast recovery (Section 3.2) — avoids the TCP global-synchronisation collapse that tail-drop queuing would otherwise cause across simultaneous large transfers.

3.5 Congestion and Flow-Control Interaction — Worked Scenario

Consider a research student's 4 GB SFTP transfer over the backbone while, on the same access switch, a lecture's video-conference and a burst of 200 students' HTTPS LMS requests occur simultaneously — a realistic peak-hour combination given R1's 5,000-user target. Without QoS, all three would compete in a single FIFO queue, with the bulk SFTP flow crowding out interactive HTTPS bursts and adding delay to video. With Table 7's DiffServ classes, video's AF41 marking guarantees a minimum bandwidth share and shorter queue, HTTPS's AF21 marking keeps page loads responsive, and SFTP is confined to the AF11/WRED bulk queue, where TCP Cubic and WRED cooperate to stay efficient without impacting the other classes. This scenario underlies the throughput/latency/jitter measurements in Section 6.1.

4. Application Protocols&Security (CO4)

4.1 Application Services Deployed

Four application-layer services are designed and simulated, covering web, file-transfer, mail and directory/monitoring categories: a campus web portal/LMS over HTTPS, research data transfer over SFTP, staff/student e-mail over IMAPS/SMTPS, and infrastructure monitoring over SNMPv3, with DNS as the shared name-resolution substrate for all of them.

4.2 DNS Design

Each campus runs a local recursive resolver in the Data Centre VLAN, caching answers and forwarding only cache misses to the university's authoritative DNS and onward to the internet hierarchy. Placing the resolver locally is what allows the R5 target (DNS resolution under 50 ms) to be met: a cached response needs only local processing plus one RTT, while a fully recursive lookup would incur multiple WAN round trips. DNSSEC validation is enabled to protect against cache-poisoning, and split-horizon DNS keeps internal service names hidden from external resolvers.

4.3 Secure Web, File Transfer and Mail

4.3.1 HTTPS / TLS 1.3

The web portal and LMS are served exclusively over HTTPS on TCP/443; plaintext HTTP on port 80 is retained only as a redirect-to-HTTPS listener via HSTS, so no session completes in cleartext. TLS 1.3 (RFC 8446) is specified over TLS 1.2 for three reasons: (1) it removes legacy cipher suites vulnerable to known attacks; (2) its 1-RTT handshake (0-RTT for resumption) reduces the latency cost of the security layer, mattering directly for the interactive AF21 traffic class; and (3) it mandates forward secrecy via ephemeral Diffie–Hellman, so a future key compromise cannot retroactively decrypt captured traffic.

4.3.2 SFTP for research data

Research data transfer uses SFTP (TCP/22) rather than legacy FTP, which sends credentials and file contents in cleartext and requires an awkward second dynamic data-channel port. SFTP multiplexes control and data over a single encrypted SSH channel, both more secure and simpler to express in the DMZ/firewall ACLs of Section 4.4.

4.3.3 Mail: IMAPS and SMTPS

Mail retrieval uses IMAP over implicit TLS (IMAPS, TCP/993) rather than plaintext IMAP or STARTTLS, avoiding any downgrade-attack window; outbound submission uses SMTPS with mandatory STARTTLS (TCP/587), with server-to-server relay authenticated and encrypted in transit.

4.4 Defence-in-Depth Security Architecture

Security is layered from the physical port up to the application session, so no single control failure exposes the network:

Layer	Control	Purpose

Layer	Control	Purpose
Physical/Access port	802.1X (EAP-TLS/PEAP) + port security (MAC limiting)	Only authenticated devices join; blocks rogue patch-in
Data-link	Native VLAN hardening, BPDU Guard, DHCP Snooping, DAI	Prevents VLAN hopping, rogue DHCP and ARP spoofing
Wireless	WPA3-Enterprise (staff/student), WPA3-SAE (IoT), isolated guest SSID	Strong per-session keys; segregates IoT/guest risk
Network	Zone-based firewall, inter-VLAN ACLs (e.g. IoT denied east-west)	Least-privilege reachability between zones (§2.4)
Transport/Session	TLS 1.3 (web/mail), SSH (management/SFTP)	Confidentiality, integrity and forward secrecy in transit
Management plane	OOB VLAN 99, SSHv2 only, SNMPv3 authPriv, centralised AAA	Protects configuration/monitoring from the data path
Monitoring	Centralised syslog, NTP-synced timestamps, SNMPv3 traps	Forensics, anomaly detection, trustworthy correlation

Table 8. Defence-in-depth control summary, layer by layer.

4.5 IoT-Specific Security Considerations

The 500 IoT sensors are the most resource-constrained and least patchable class of device on campus, so they are treated as an explicitly lower-trust zone: isolated on VLAN 50 with default-deny east-west ACLs to every other VLAN except a single telemetry-ingest endpoint, rate-limited at the access port, and using a lightweight authentication profile appropriate to constrained hardware rather than the full EAP-TLS stack used for staff laptops.

4.6 Threat Model Summary

Threat	Primary Mitigation(s)
Eavesdropping on web/mail/file-transfer traffic	TLS 1.3 (HTTPS/IMAPS/SMTPS), SSH (SFTP) — §4.3
Rogue access point / rogue DHCP	WPA3-Enterprise, DHCP Snooping, WLC intrusion detection
ARP spoofing / on-path MITM within a VLAN	Dynamic ARP Inspection, DHCP Snooping binding table
Compromised IoT device pivoting to core systems	VLAN isolation, default-deny ACLs, rate-limiting (§4.5)
Unauthorised device management access	AAA (TACACS+/RADIUS), SSH-only, SNMPv3 authPriv, OOB VLAN
Broadcast storm / Layer-2 loop from mis-patched cable	BPDU Guard, PortFast on true edge ports (§1.4)
DNS cache poisoning	DNSSEC validation at the recursive resolver (§4.2)

Table 9. Threat-to-mitigation mapping.

5. Use of Modern Tools: Packet Tracer, Wireshark, GitHub&Simulation Evidence (CO1–CO3)

5.1 Note on Evidence Placeholders

Figures and capture excerpts below are marked [INSERT SCREENSHOT]/[INSERT CAPTURE] because they must come from the team's own Packet Tracer (.pkt) and Wireshark (.pcapng) captures, run on this design, to satisfy the requirement for genuine, reproducible simulation evidence and to avoid the academic-misconduct risk flagged for unacknowledged AI-generated evidence. Section 5.5 gives the exact build checklist.

5.2 Cisco Packet Tracer Topology

The topology implements the hierarchical design (Section 1.4) and the addressing plan (Section 2). The device inventory below sizes to represent all five zones within Packet Tracer's practical device-count limits for a single .pkt file.

Device Role	Model	Count	Key Configuration
Core switch	Catalyst 3650/3850 (L3)	2	HSRP active/standby, OSPF Area 0, 10 Gbps EtherChannel
Distribution switch	Catalyst 3650 (L3)	5 (1 per zone)	OSPF ABR, inter-VLAN SVIs, DHCP relay, SDN agent
Access switch	Catalyst 2960	10 (2 per zone)	802.1X, port security, DHCP snooping
Wireless LAN Controller	Cisco 3504 WLC	1	AP management, WPA3-Enterprise SSIDs, band steering
Access Points	802.11ax-capable	12	2–3 per teaching/residential building, OFDMA enabled
Edge router/firewall	ISR 4321 with ZBFW	1	Default route to internet, DMZ zone, outbound NAT
Servers	Generic Server	5	DNS/DHCP/Web/Mail/File, Data Centre VLAN 60
End devices	PC/Phone/IoT sensor	≈30	Sample from each VLAN for connectivity/QoS testing
SDN controller node	PT SDN controller	1	Programs backup paths per Section 2.5.3

Table 10. Packet Tracer device inventory for the campus topology.

[INSERT SCREENSHOT: full-topology view of the .pkt file, showing all five zones, the core pair, and wired/wireless/trunk link colour-coding.]

5.3 Validation Checklist Performed in Packet Tracer

- End-to-end ping and traceroute between a Teaching-zone wireless client and a Data-Centre server, confirming inter-VLAN routing and OSPF reachability.
- show ip route and show ip ospf neighbor on a distribution switch, confirming the summarised /19 routes (Section 2.5.2).

- Simulated link failure between a distribution switch and its primary core uplink, timed against the R6 target.
- show spanning-tree confirming Rapid PVST+ root placement and BPDU Guard err-disabling a port when a rogue switch is simulated.
- A blocked ping from an IoT-VLAN device to the Staff-Wired VLAN, demonstrating the default-deny ACL (Section 4.5).
- A successful HTTPS session versus a blocked/redirected plain-HTTP attempt, demonstrating the HSTS behaviour (Section 4.3.1).
- DHCP lease acquisition captured for clients in two different VLANs, confirming the relay/scope design (Section 2.5.4).

5.4 Wireshark Capture Plan

Captures are taken at vantage points covering control-plane, data-plane and security-relevant traffic, saved as separate .pcapng files named per Section 5.6.

Capture Point	Filter Focus	What It Demonstrates
Distribution-switch uplink	ospf, bfd	Hello/DBD/LSAck exchange, Type-3 LSA summarisation
Access-port mirror (Teaching-A, VLAN 20)	tcp.flags.syn==1, tls.handshake	TCP 3-way handshake followed by the TLS 1.3 handshake
Access-port mirror, same client	http	Port-80 request showing only the 301/HSTS redirect
Data-Centre VLAN mirror	dhcp, dns	DHCP lease sequence, recursive DNS timing (R5 target)
Core inter-switch link (induced failure)	icmp, ospf	Packet loss/gap during failover, vs 2-RTT target (R6)
IoT-VLAN mirror	ip.addr==10.50.x.x	Confirms no IoT traffic reaches Staff/Server VLANs

Table 11. Wireshark capture points and analysis targets.

[INSERT CAPTURE: annotated TCP three-way handshake + TLS 1.3 handshake, with RTT deltas visible in the Wireshark time column.]

5.5 Build Checklist for the Team

1. Lay out the five-zone physical topology in Packet Tracer per Table 10.
2. Apply the VLSM addressing from Section 2 exactly as tabulated.
3. Configure VLANs, trunking, SVIs and HSRP per Sections 1.4 and 2.4.
4. Configure OSPF multi-area routing and BFD per Section 2.5; verify with show ip ospf neighbor/show ip route.
5. Configure DHCP relay, the DiffServ QoS policy (Table 7 DSCP values), and the security controls (Table 8).

6. Run the Section 5.3 validation checklist in Simulation mode, capturing screenshots at each step.
7. Save the Section 5.4 capture points as .pcapng files, annotating at least one screenshot per point.
8. Replace every placeholder with actual evidence and record measured values in Section 6's results tables before submission.

5.6 GitHub Repository Structure

All configuration, topology and documentation artefacts are version-controlled in a single GitHub repository (mandatory deliverable).

Path	Contents
/README.md	Setup instructions, topology summary, contribution table
/packet-tracer/campus-network.pkt	The complete working topology file
/configs/	Per-device running-configs exported as .txt
/wireshark/	Saved .pcapng captures, one per vantage point (Table 11)
/diagrams/	Logical and physical topology diagrams
/docs/technical-report.pdf	This report, exported as the final submission
/docs/addressing-plan.xlsx	The VLSM/VLAN tables from Section 2 in spreadsheet form
/.gitignore, /LICENSE	Standard repository hygiene files

Table 12. Recommended GitHub repository layout.

Commit history is graded evidence of individual contribution; each member should commit their own work under their own GitHub account, with messages referencing the section supported (e.g., "Add OSPF Area 20 config — supports Sec. 2.5.1").

6. Results, Trade-off Analysis&Engineering Justification (CO2/CO3)

6.1 Analytically Expected Performance

Before presenting simulated results, the team derives analytically expected values, both to sanity-check Packet Tracer measurements and because quantitative claims must be backed by tool evidence or engineering calculation.

6.1.1 Bandwidth-delay product (backbone sizing)

For a 1 Gbps core link with an intra-campus RTT of 2 ms: BDP = 1×10^9 bps × 0.002 s ÷ 8 = 250,000 bytes (≈244 KiB). This is the minimum TCP receive-window/buffer size needed to keep the backbone link fully utilised by a single bulk flow (e.g., the SFTP transfer in Section 3.5); modern OS TCP window scaling comfortably exceeds it without manual tuning.

6.1.2 Expected TCP goodput under wireless loss (Mathis formula)

Using Throughput ≈ MSS ÷ (RTT × $\sqrt{p}$), with MSS = 1460 B = 11,680 bits and RTT = 2 ms: at the R4 boundary loss rate p = 1%, throughput ≈ 11,680 ÷ (0.002 × $\sqrt{0.01}$) ≈ 58.4 Mbps per flow — comfortably above a single student's LMS session needs. At a degraded p = 5% (pre-upgrade collision-driven loss), throughput ≈ 26.1 Mbps per flow — over a 2× reduction, quantifying why the OFDMA/RTS-CTS improvements (Section 1.3.2) directly protect application-layer throughput.

6.2 Target vs Simulated Results

Table 13 restates the performance requirements alongside the analytically expected value and a column for the team's own measured value from Packet Tracer/Wireshark evidence; the Measured column must be completed from the team's own captures before final submission.

Metric	Target (R#)	Analytically Expected	Measured
Wireless-zone packet loss	<1% (R4)	~0.3–0.9% with OFDMA + RTS/CTS	[insert from Wireshark]
Local DNS resolution time	< 50 ms (R5)	< 5 ms cached, 15–40 ms recursive	[insert dns.time]
TCP goodput recovery after single loss	≤ 2 RTT (R6)	1–2 RTT via fast retransmit/recovery	[insert tcp.analysis.ack_rtt]
Core backbone utilisation (peak)	well within 1 Gbps	<40% under modelled peak load	[insert from Simulation stats]
Single-link failover time (OSPF+BFD)	sub-second, then ≤2 RTT	≈150–400 ms detection, SPF <100 ms	[insert from induced-failure test]
VoIP one-way delay / jitter	< 30 ms (Table 7)	10–25 ms LAN, priority-queued	[insert from RTP capture]

Table 13. Target vs analytically-expected vs measured performance (Measured column to be completed by the team).

6.3 Trade-off Analysis

6.3.1 Link-state (OSPF) vs distance-vector (EIGRP/RIP)

Criterion	OSPF (chosen)	EIGRP	RIP
Convergence	Fast — event-driven SPF, faster with BFD	Fast via DUAL, proprietary tuning needed	Slow — 30 s periodic updates, up to 180 s timeout
Scalability	Areas bound SPF scope; scales well	Good, less standardised summarisation	Poor — 15-hop limit, count-to-infinity risk
Standardisation	Open IETF standard (RFC 2328)	Cisco-originated, partially opened	Open but obsolete for enterprise use
Fit for this campus	Best — matches multi-zone, multi-vendor future	Viable, used only as legacy comparison	Rejected outright

Table 14. Interior routing protocol trade-off (supports Section 2.5.1).

The legacy Residential-zone EIGRP segment gives the team an empirical convergence comparison: an induced failure on the OSPF/BFD segment should show detection-to-reroute inside a few hundred milliseconds, while the same failure on the EIGRP segment (without BFD) should take noticeably longer under DUAL's default hold-time behaviour — a stronger results demonstration than either protocol's textbook description alone.

6.3.2 Wi-Fi 6 (802.11ax) vs Wi-Fi 5 (802.11ac)

802.11ax was chosen primarily for OFDMA, letting one AP serve many simultaneous small transmissions (typical of a lecture hall of laptops making short HTTPS requests) far more efficiently than 802.11ac's one-station-at-a-time allocation, directly addressing R1/R4. The trade-off is client-side: benefits are only fully realised as client devices themselves adopt 802.11ax, a phased-adoption caveat rather than a design flaw.

6.3.3 VLAN-segmented vs flat network

A flat, single-broadcast-domain design was rejected because at 5,000+ devices the broadcast/ARP overhead alone would consume meaningful access-link capacity, and it offers no basis for IoT isolation (Section 4.5). The administrative cost of segmentation is judged minor since inter-VLAN routing is effectively free at line rate on modern L3 switch silicon.

6.3.4 Centralised SDN overlay vs pure distributed routing

OSPF alone is correct but reactive, rerouting only after detecting failure. The SDN overlay trades a small amount of operational complexity — a redundantly deployed controller — for pre-computed backup paths with near-zero reconvergence delay and policy-based traffic steering, judged worthwhile given R6's tight recovery target.

6.3.5 Security depth vs performance/complexity overhead

Every control in Table 8 adds some overhead — 802.1X adds authentication latency at association, TLS 1.3 adds at most one round trip, DHCP Snooping/DAI add per-packet inspection. None individually threaten the Table 13 targets (TLS 1.3's handshake cost is sub-10 ms and one-time per session), so the design retains the full defence-in-depth stack given the university's data-sensitivity profile.

6.4 Limitations and Assumptions

- The "Measured" values in Table 13 must be populated from the team's own Packet Tracer/Wireshark run; this report supplies the design and analytically expected values but was not itself run against a live simulation.
- IPv6 is planned for but not implemented in the current topology (Section 2.6).
- The SDN overlay is represented at Packet Tracer's SDN-emulation fidelity rather than a full OpenFlow controller.
- Wireless propagation modelling is treated at a planning level (AP counts, OFDMA, RTS/CTS policy) rather than full RF simulation.
- All numeric targets are taken from the assignment brief; where silent, an industry-standard value is used and cited in Section 10.

7. Broader Considerations (Sustainability, Security/Privacy, Cost, Accessibility, Professional Responsibility, SDG)

7.1 Sustainability and Energy Efficiency

802.11ax's Target Wake Time is used on the IoT SSID (Section 4.5) so battery-powered sensors can negotiate scheduled wake/sleep cycles instead of staying radio-active continuously, extending device battery life across the 500-device deployment. Route summarisation (Section 2.5.2) keeps core control-plane CPU/power draw low as the campus scales, and consolidating DNS, DHCP and mail in a single Data-Centre VLAN lets the university concentrate cooling/power provisioning in one well-managed facility.

7.2 Security and Privacy

The defence-in-depth architecture (Section 4.4) and mandatory TLS 1.3/SFTP/IMAPS posture (Section 4.3) are, in effect, this project's privacy design: no traffic traverses the network in cleartext by default, IoT devices are isolated so a compromise cannot become a foothold into student-record or research systems, and centralised AAA/logging gives an auditable record for incident response and data-protection compliance. QoS marking inspects only header fields (DSCP, port numbers), never payload content — a deliberate choice preserving confidentiality even from the network's own management plane.

7.3 Cost Considerations

The hierarchical design reuses a small number of switch models across many identical roles (Table 10) rather than bespoke per-building hardware, keeping spares, training and support costs down; centralising DHCP/DNS/mail avoids per-building licensing costs. The main added cost relative to a minimal flat network is the distribution-layer L3 tier and WLC/AP density needed for R1/R4 — judged justified against the recurring productivity cost of the pre-upgrade collision/loss problems.

7.4 Accessibility and Reliability

Redundant core switches (HSRP), dual-homed distribution uplinks, and the SDN backup-path overlay keep the network available with no single link or device failure taking a whole zone offline — an accessibility concern given students who rely on campus Wi-Fi and lack reliable home internet are disproportionately affected by outages. The captive-portal guest SSID also extends basic connectivity to visitors and campus events.

7.5 Professional Responsibility and Ethics

The design documents its own limitations plainly (Section 6.4) rather than overstating simulated results as production-measured data, and the GitHub commit-history practice (Section 5.6) makes individual contribution verifiable rather than asserted, consistent with the academic-integrity requirement that AI-assisted content be acknowledged. Security decisions default to the more conservative option wherever there was a choice (TLS 1.3 over 1.2, SFTP over FTP, WPA3 over WPA2).

7.6 Sustainable Development Goal (SDG) Mapping

SDG	Linkage
SDG 4 — Quality Education	Reliable, low-latency access to the LMS and research systems (R1, R4–R6) removes the connectivity barrier to consistent teaching identified in Section 1.1.
SDG 9 — Industry, Innovation and Infrastructure	The IoT sensor network, SDN-controlled segment and Wi-Fi 6 deployment are exactly the resilient, innovative infrastructure SDG 9 targets, built to scale as the university grows.
SDG 11 — Sustainable Cities and Communities	The energy-monitoring IoT deployment and TWT/route-summarisation efficiency measures (Section 7.1) support the university's role as a small-scale "smart campus" community.

Table 15. SDG mapping (per SDG 4, 9, 11 named in the assignment brief).

7.7 Summary of Findings, Limitations and Possible Improvements

Findings. The redesigned campus network replaces a flat, collision-prone, slowly-converging network with a hierarchical, VLSM-addressed, OSPF-routed, DiffServ-QoS'd, TLS-secured design that analytically meets every numeric target in the brief (Table 13), with a reproducible Packet Tracer/Wireshark methodology to confirm those figures empirically.

Limitations. As listed in Section 6.4: measured results depend on the team completing the evidence capture; IPv6 and full RF-propagation modelling are out of scope; and the SDN overlay is represented at Packet Tracer's level of fidelity rather than a full OpenFlow implementation.

- Phase in IPv6 dual-stack per zone using the reserved ULA plan (Section 2.6).
- Extend the SDN overlay's flow-rule policy to a full multi-constraint optimiser as device counts grow.
- Add a formal RF site survey once physical APs are installed, replacing planning-level AP density assumptions with measured data.
- Introduce network-wide 802.1X certificate-based authentication for IoT devices as constrained-device EAP methods mature.

8. Technical Documentation, Team Contribution&Reflection (CO1–CO3, Professional)

8.1 Documentation Approach

This report is structured to mirror the assessment rubric one-to-one (Sections 1–7 map directly to the eight rubric rows), so graders and teammates can trace every mark to a specific, labelled section rather than searching prose for scattered evidence. Addressing, VLAN and QoS tables are kept internally consistent (the same VLAN numbers, DSCP values and IP ranges recur in Sections 2–5) so the GitHub artefacts in Section 5.6 can be checked against this document line-by-line.

8.2 Suggested Team Roles

A four-member team maps naturally onto the rubric's own groupings; the suggested division is a starting point the team should adjust to actual strengths, then record accurately in Table 17.

Role	Primary Rubric Areas	Typical Responsibilities
Network Design Lead	Problem understanding, IP addressing, routing (§1–2)	Topology design, VLSM/VLAN plan, OSPF/SDN configuration
Transport&Security Lead	Transport/QoS, application/security (§3–4)	QoS policy, TLS/SFTP/IMAPS services, firewall/ACL design
Simulation&Validation Lead	Tools&evidence, results (§5–6)	Packet Tracer build, Wireshark captures, measurement/analysis
Documentation&Integration Lead	Broader considerations, documentation, GitHub (§7–8)	Report assembly, repository hygiene, SDG analysis, final QA

Table 16. Suggested role allocation — replace with the team's actual assignment.

8.3 Individual Contribution Statement (Template)

The assignment requires each member's contribution to be specific, signed and honest. Complete this with real names, real GitHub usernames and factual descriptions — vague statements ("helped with everything") score poorly and are not a truthful record.

Member (Name, Roll No.)	GitHub Username	Sections/Modules Owned	Signature
[Member 1]	[username]	[e.g. §1–2; OSPF and VLSM configuration]	[signature]
[Member 2]	[username]	[e.g. §3–4; QoS and security configuration]	[signature]
[Member 3]	[username]	[e.g. §5; Packet Tracer build and captures]	[signature]
[Member 4]	[username]	[e.g. §7–8; report integration, GitHub]	[signature]

Table 17. Individual contribution statement — to be completed and signed by all members.

9. Team Reflection

The assignment brief asks what each member learned beyond classroom teaching, what challenges the team faced using Packet Tracer, Wireshark and GitHub, and how more time or resources would have changed the solution. Each member should write two to three honest sentences under their own name for each prompt below, drawing on their own experience building and testing the topology (Section 5).

9.1 Prompts for Each Member

- **Beyond the classroom:** What did building this topology teach you that lectures/textbook reading did not?
- **Packet Tracer challenges:** Which feature (SDN/OpenFlow emulation, QoS policy-maps, WLC configuration, Simulation-mode tracing) was hardest to get working, and how was it resolved?
- **Wireshark challenges:** Was capturing at the intended vantage points (Table 11) straightforward, and did anything in the capture change the team's understanding of the design?
- **GitHub/collaboration challenges:** How did the team handle the non-mergeable .pkt file and keep commit history meaningful?
- **With more time or resources:** What would the team have added or done differently — a full RF site survey, IPv6 dual-stack, a larger endpoint sample, or load-testing closer to the full 5,000-user target?

9.2 Space for Recorded Reflections

[Member 1 Name] — Beyond the classroom: [to be completed]. Tool challenges: [to be completed]. With more time: [to be completed].

[Member 2 Name] — Beyond the classroom: [to be completed]. Tool challenges: [to be completed]. With more time: [to be completed].

[Member 3 Name] — Beyond the classroom: [to be completed]. Tool challenges: [to be completed]. With more time: [to be completed].

[Member 4 Name] — Beyond the classroom: [to be completed]. Tool challenges: [to be completed]. With more time: [to be completed].

10. References

IEEE 802.11ax-2021, IEEE Standard for Information Technology — Telecommunications and Information Exchange between Systems — Local and Metropolitan Area Networks — Amendment for Enhancements for High-Efficiency WLAN.

RFC 791 — Internet Protocol, IETF, 1981.

RFC 793 — Transmission Control Protocol, IETF, 1981.

RFC 1918 — Address Allocation for Private Internets, IETF, 1996.

RFC 2328 — OSPF Version 2, IETF, 1998.

RFC 3550 — RTP: A Transport Protocol for Real-Time Applications, IETF, 2003.

RFC 6928 — Increasing TCP's Initial Window, IETF, 2013.

RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3, IETF, 2018.

Mathis, M., Semke, J., Mahdavi, J. and Ott, T., "The Macroscopic Behavior of the TCP Congestion Avoidance Algorithm," ACM SIGCOMM Computer Communication Review, 1997.

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Tanenbaum, A. S. and Wetherall, D. J., Computer Networks, Pearson, latest edition.

Cisco Systems, Cisco Packet Tracer — Official Documentation, Cisco Networking Academy.

Wireshark Foundation, Wireshark User's Guide, wireshark.org/docs.

United Nations, The 17 Sustainable Development Goals, sdgs.un.org.

Cisco Systems, Quality of Service (QoS) DiffServ Configuration Guide, Cisco Networking Academy.

10.1 Use of AI Assistance

Per the assignment's instruction that AI-assisted work be properly cited, the team should record here, factually, which parts of the submission (if any) used AI assistance and how the output was reviewed, verified and adapted — for example, drafting/structuring assistance for this report template, with all technical design choices, addressing calculations and simulation evidence independently checked, executed and validated by the team in Packet Tracer and Wireshark. Un-acknowledged AI-generated content is treated by the department as academic misconduct, so this disclosure should be completed honestly and specifically rather than omitted.

Appendix A — Glossary of Abbreviations

Term	Meaning
ABR	Area Border Router (OSPF)
ACL	Access Control List
BDP	Bandwidth-Delay Product
BFD	Bidirectional Forwarding Detection
CRC	Cyclic Redundancy Check
DHCP	Dynamic Host Configuration Protocol
DiffServ	Differentiated Services
DMZ	Demilitarised Zone (network segment)
DSCP	Differentiated Services Code Point
FCS	Frame Check Sequence
HSRP	Hot Standby Router Protocol
MSS	Maximum Segment Size
OFDMA	Orthogonal Frequency-Division Multiple Access
OSPF	Open Shortest Path First
QoS	Quality of Service
RTT	Round-Trip Time
SDN	Software-Defined Networking
SVI	Switch Virtual Interface
TLS	Transport Layer Security
TWT	Target Wake Time (802.11ax)
VLSM	Variable-Length Subnet Masking
WRED	Weighted Random Early Detection

Appendix B — Additional Worked VLSM Example: Residential Hall A (10.40.0.0/22)

Provided as a second worked example demonstrating the lighter four-VLAN template referenced in Section 2.3 for Residential and Administration buildings.

VLAN / Segment	Hosts Needed	Mask	Network	Usable Range
20 – Resident Wireless	480	/23 (510)	10.40.0.0	10.40.0.1 – 10.40.1.254
10 – Wired (staff/common-room)	25	/27 (30)	10.40.2.0	10.40.2.1 – 10.40.2.30
99 – Management	6	/29 (6)	10.40.2.32	10.40.2.33 – 10.40.2.38
P2P – Uplink to Distribution	2	/30 (2)	10.40.2.40	10.40.2.41 – 10.40.2.42

Table B.1. VLSM subnetting for Residential-A (556 of 1,024 addresses used).

Appendix C — Illustrative Device Configuration Skeleton

The following is a generic, standards-based configuration skeleton (Cisco IOS-style syntax, as used by Packet Tracer) illustrating how the design in Sections 2–4 translates into device configuration. It is a teaching skeleton for the team to adapt to each specific device, not a copy-paste final configuration — actual configs exported from the working .pkt file belong in the /configs/ folder of the GitHub repository per Section 5.6.

```
interface Vlan20
 description Student-Wireless
 ip address 10.10.0.1 255.255.254.0
 ip helper-address 10.60.0.10
!
router ospf 1
 router-id 10.10.255.1
 area 10 range 10.10.0.0 255.255.224.0
 network 10.10.0.0 0.0.31.255 area 10
 bfd all-interfaces
!
class-map match-all VOICE
 match dscp ef
policy-map CAMPUS-QOS
 class VOICE
  priority percent 10
 class BULK-DATA
  bandwidth remaining percent 30
  random-detect dscp-based
```

Appendix D — Deliverables Checklist (from Assignment Brief)

- Technical Report (this document), 25–30 pages excluding appendices — structured per the CO-mapped rubric.
- Cisco Packet Tracer .pkt file (or GNS3/Mininet-equivalent) of the complete working topology.
- Wireshark .pcapng capture files demonstrating ARP, ICMP, TCP, DNS and key protocol exchanges.
- GitHub repository link (mandatory) containing the .pkt file, configuration text files, capture/screenshot evidence, topology diagrams, README with setup instructions, and full commit history from every member.
- Team contribution statement, signed by all members, indicating individual responsibilities and tools used per person.
- Optional but encouraged: a short (≤5 minute) recorded screen demonstration of the Packet Tracer topology and key Wireshark captures.